MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, AND-OR-INVERT, DIGITAL, KEY CODE EBA

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic AND-OR-INVERT gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.6 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EBA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/170-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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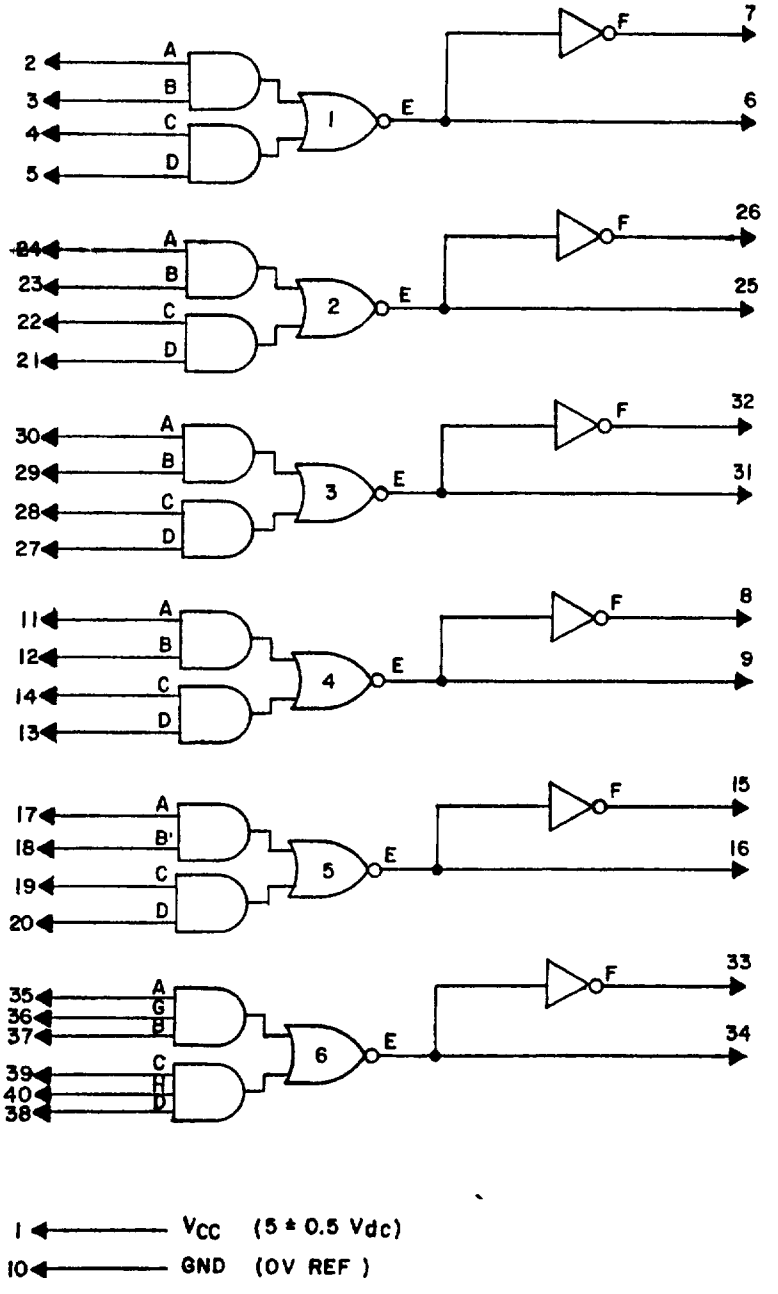


FIGURE 1. EBA module function diagram, AND-OR-INVERT gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirement 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low-loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B26	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
C	Isolation, frame	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D18	Output voltage, low level	VI	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
E1-E18	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F26	Input current, low level	VIII	X	50	200	50	200	50	200	50	200	µA
G1-G26	Input current, high level	IX	X	0	10	0	10	0	10	0	10	µA
H1-H12	Output short circuit current	X	X	3	15	3	15	3	15	3	15	mA
J1	Power supply current, low level	XI	S	0.5	5.1	0.5	5.1	0.5	5.1	0.5	5.1	mA
J2	Power supply current, high level	XI	S	0.2	5.5	0.2	5.5	0.2	5.5	0.2	5.5	mA
K1-K12	Transition time (t_{TLH}), C_L = 40 pF	XII	S	5	60	5	60	5	60	5	60	ns
K13-K24	Transition time (t_{TLH}), C_L = 140 pF	XII	Q	10	100	10	100	10	100	10	100	ns
L1-L12	Trarsition time (t_{THL}), C_L = 40 pF	XIII	S	2	30	2	30	2	30	2	30	ns
L13-L24	Transition time (t_{THL}), C_L = 140 pF	XIII	Q	5	60	5	60	5	60	5	60	ns
M1-M12	Propagation delay time (t_{PLH}), C_L = 40 pF	XIV	S	10	90	10	90	10	100	10	100	ns
M13-M24	Propagation delay time (t_{PLH}), C_L = 140 pF	XIV	Q	10	120	10	120	10	130	10	130	ns
N1-N12	Propagation delay time (t_{PLH}), C_L = 40 pF	XIV	S	10	150	10	150	10	165	10	165	ns
N13-N24	Propagation delay time (t_{PLH}), C_L = 140 pF	XIV	Q	10	180	10	180	10	195	10	195	ns
P1-P12	Propagation delay time (t_{PHL}), C_L = 40 pF	XV	S	10	60	10	60	10	70	10	70	ns

See footnotes at end of table.

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TABLE I . Module electrical requirements -Continued.

Test condition	Parameter	Ref test Table	Test type 1/-3/	Acceptance requirement 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P13-P24	Propagation delay time (t_{PHL}), C_L = 140 pF	XV	Q	10	90	10	90	10	100	10	100	ns
R1-R12	Propagation delay time (t_{PHL}), C_L = 40 pF	XV	S	10	120	10	120	10	130	10	130	ns
R13-R24	Propagation delay time (t_{PHL}), C_L = 140 pF	XV	Q	10	150	10	150	10	160	10	160	ns
S1-S12	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
T1-T12	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XVIII	Q	0	300	0	300	0	300	0	300	mV
V	Power supply filter capacitance	XIX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
W1-W26	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

NOTES:
1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through R and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/170-1 supersedes part number M28787/170. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

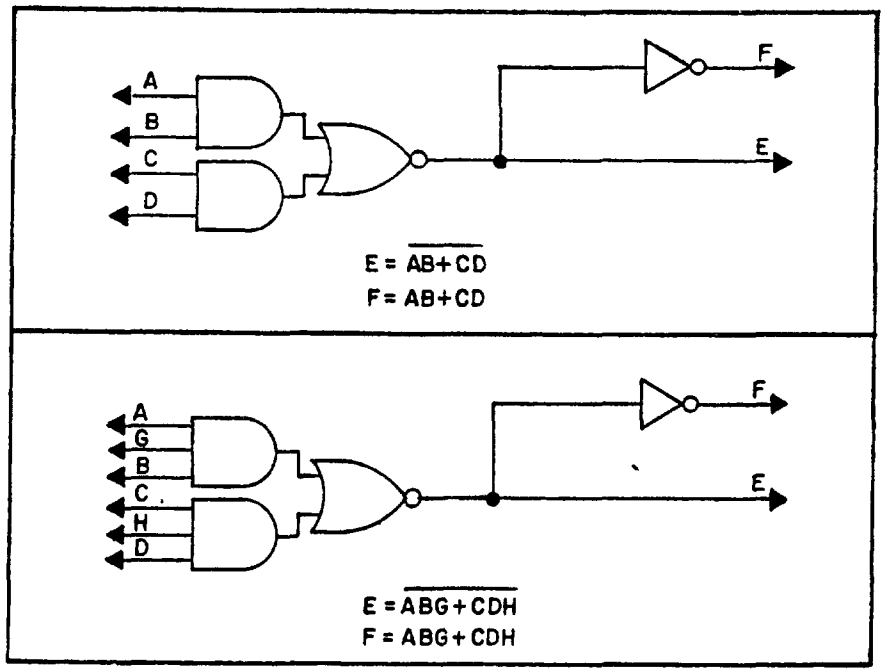
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Maximum power dissipation	
All outputs at low level	28 mW (outputs not loaded)
All outputs at high level	30 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current	
Low level (V_{IL} = 0.0 V)	-0.2 mA maximum
(V_{IL} = 0.3 V)	-0.18 mA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	100 µA maximum
Output voltage	
Low level (I_{OL} = 2.0 mA)	0.3 V maximum
High level (I_{OH} = -100 µA)	2.4 V maximum
Output drive current maximum capability	
Low level (V_{OL} = 0.3 V)	2.0 mA (F outputs)
	1.8 mA (E outputs)
High level (V_{OH} = 2.4 V)	-100 µA (F outputs)
	-90 µA (E outputs)

6.5 Module function. This module contains five independent 2-2-input and one 3-3-input AND-OR-INVERT gates.

6.5.1 Input to output relationship. Each AND-OR-INVERT gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.



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6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

The circuit groups which can be paralleled are:

(1) Circuits 1 and 2.

(2) Circuits 3 and 4.

(3) Circuits 5 and 6.

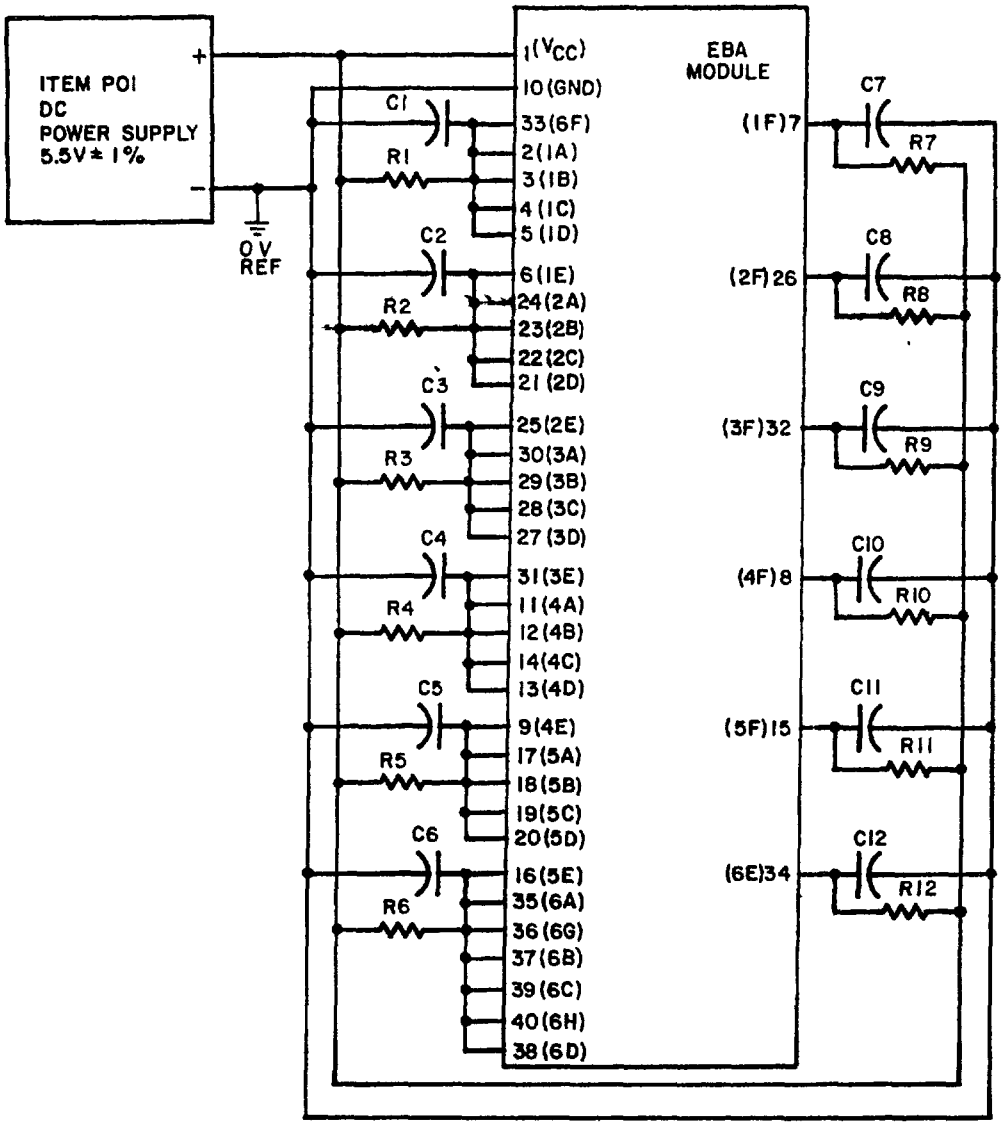
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
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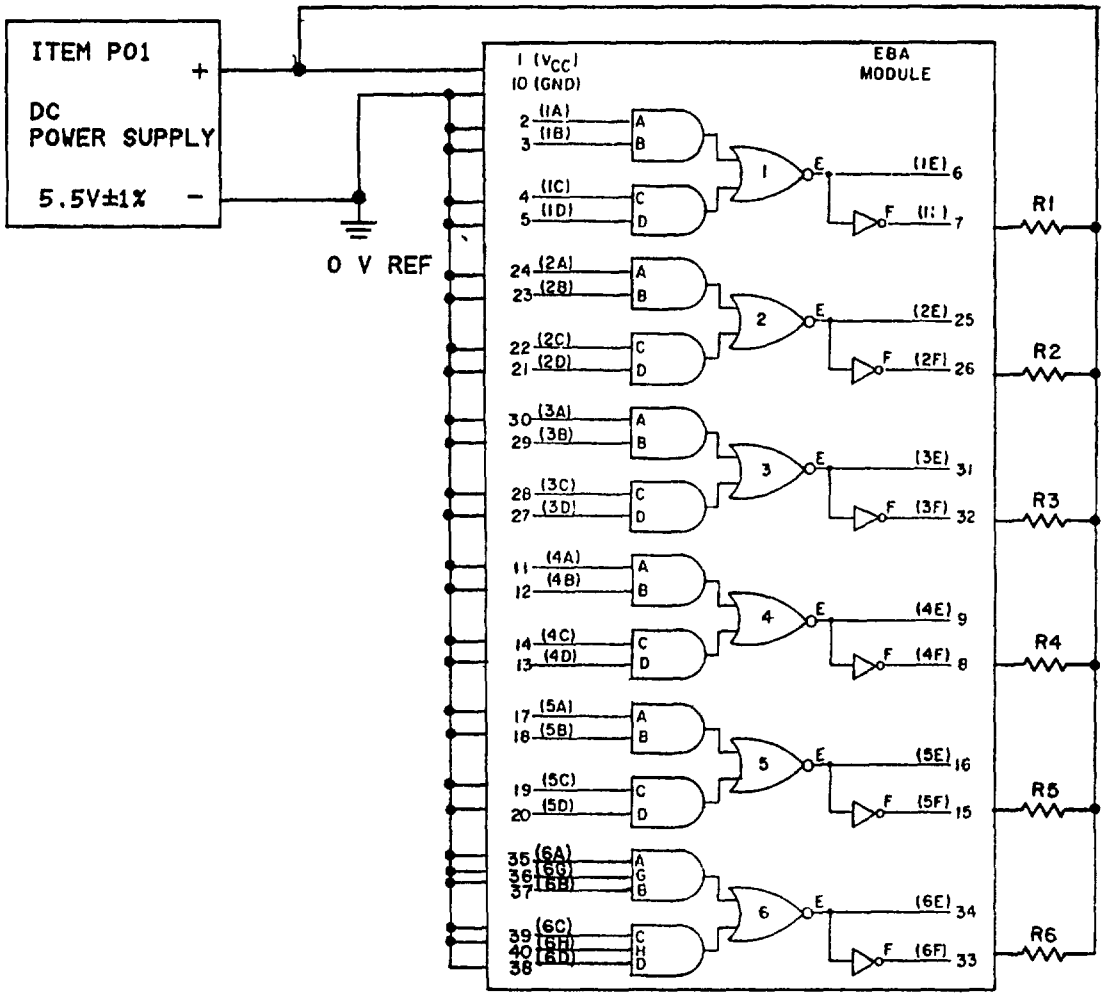
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NOTES:
1. R1-R12 = 2.75 kΩ ±1%, 1/8 W.
2. C1-C12 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R6 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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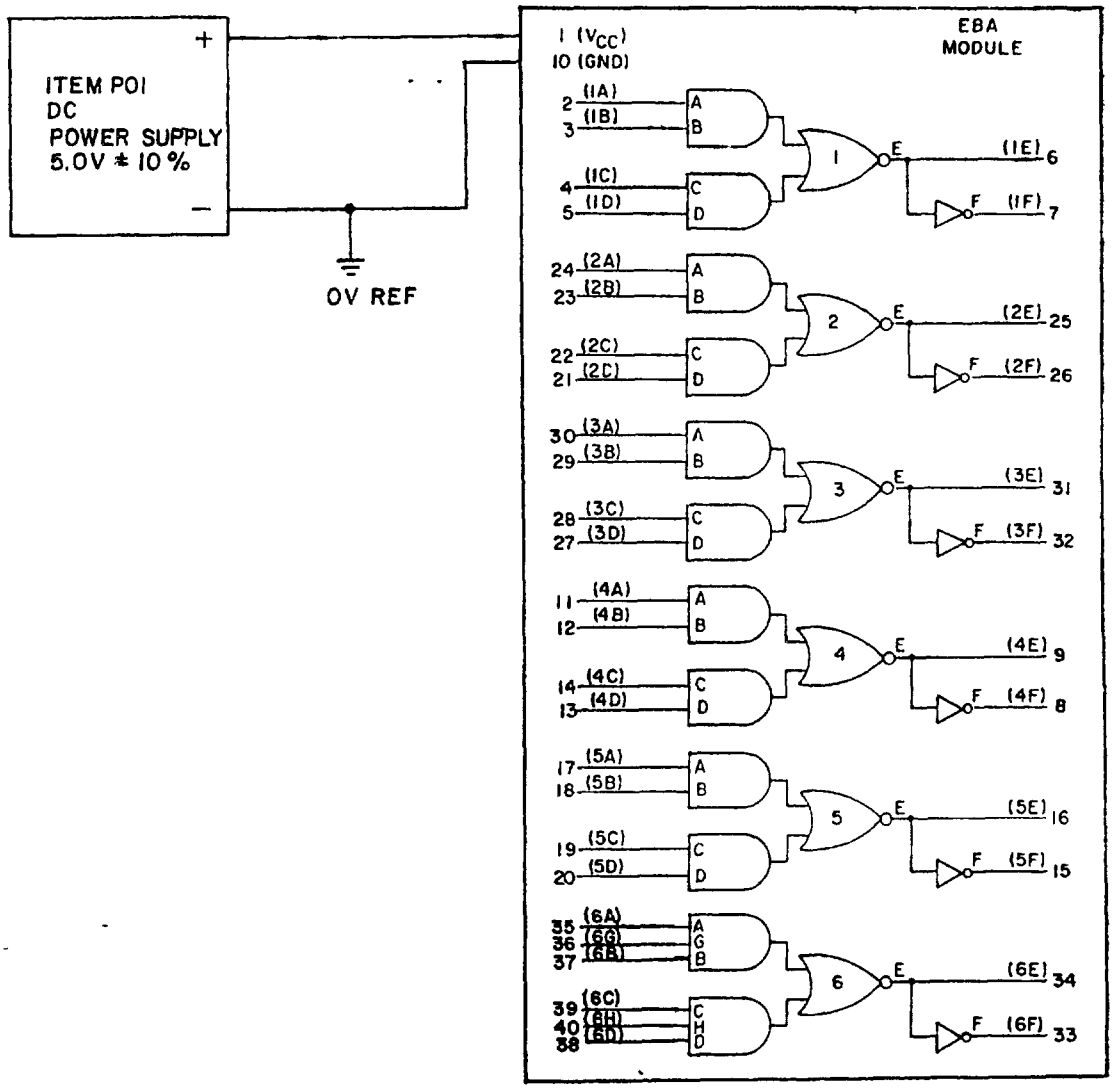


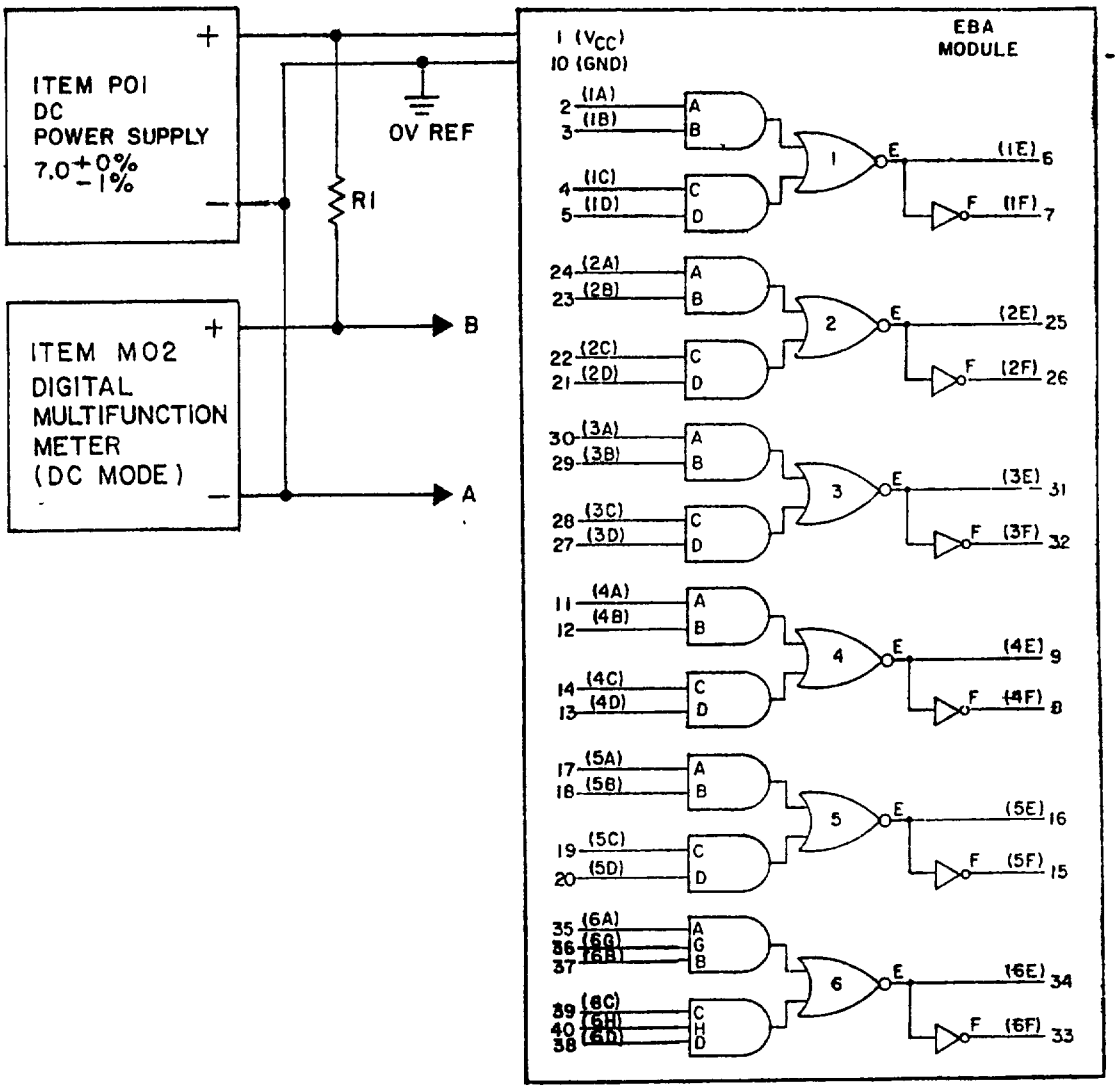
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	1E V_{OUT} (low loaded)	Open	6	0.0	0.6	V	5
A2	2E	↓	25	↓	↓	↓	↓
A3	3E		31				
A4	4E		9				
A5	5E		16				
A6	6E		34				
A7	1F	3, 4	7				
A8	2F	22, 24	26				
A9	3F	30, 28	32				
A10	4F	13, 12	8				
A11	5F	19, 17	15				
A12	6F	39, 37	33				

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NOTES:
1. R1 = 2.75 kΩ ±1%, 1/8 W.
2. Measurements shall be recorded after point "B" has been connected to the appropriate module pin for at least 5 seconds.

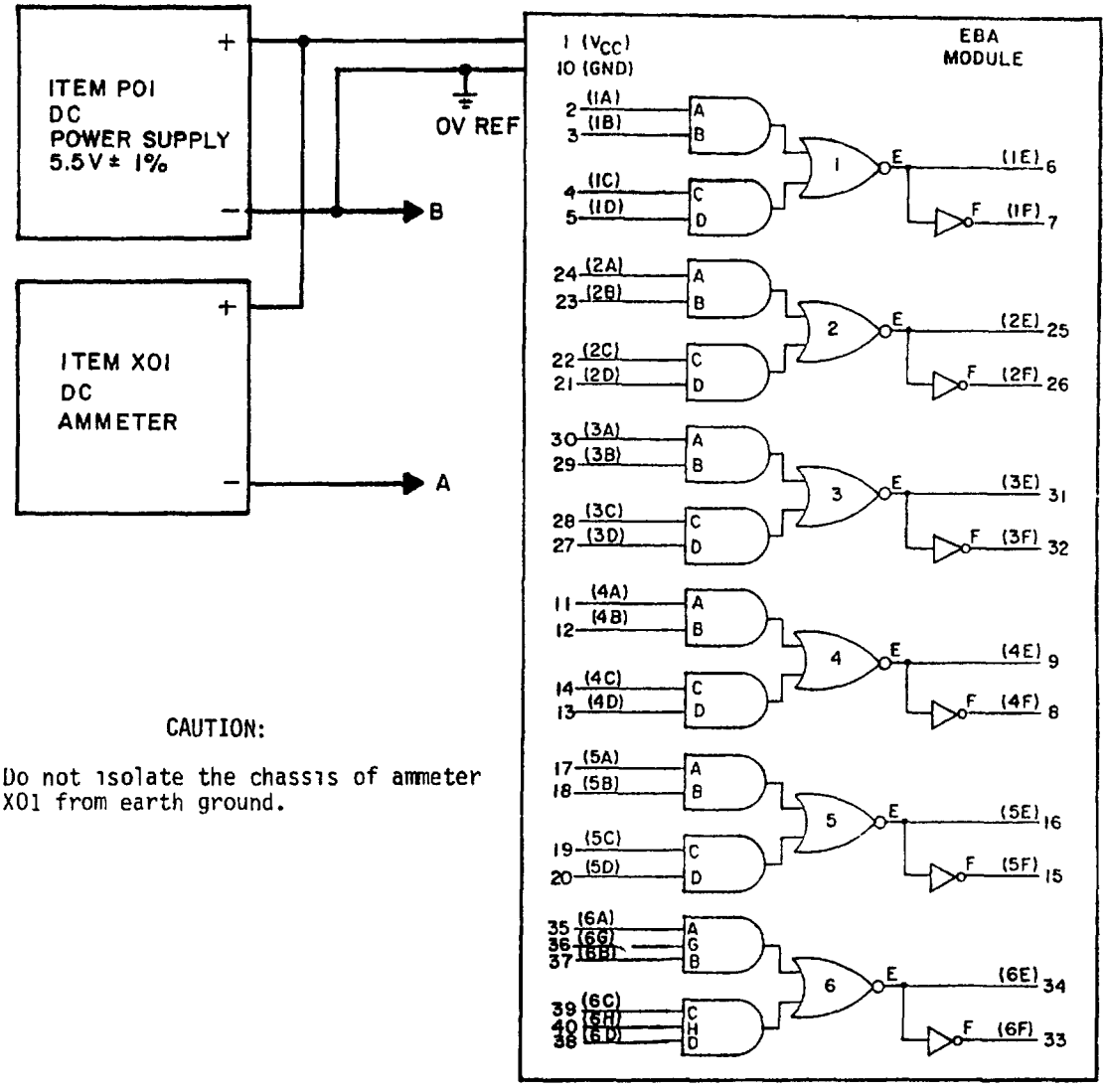
FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input breakdown current	2	3	0	100	µA	6
B2	1B	3	2				
B3	1C	4	5				
B4	1D	5	4				
B5	2A	24	23				
B6	2B	23	24				
B7	2C	22	21				
B8	2D	21	22				
B9	3A	30	29				
B10	3B	29	30				
B11	3C	28	27				
B12	3D	27	28				
B13	4B	12	11				
B14	4A	11	12				
B15	4C	14	13				
B16	4D	13	14				
B17	5A	17	18				
B18	5B	18	17				
B19	5D	20	19				
B20	5C	19	20				
B21	6A	35	36, 37				
B22	6G	36	35, 37				
B23	6B	37	35, 36				
B24	6D	38	39, 40				
B25	6C	39	38, 40				
B26	6H	40	38, 39				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

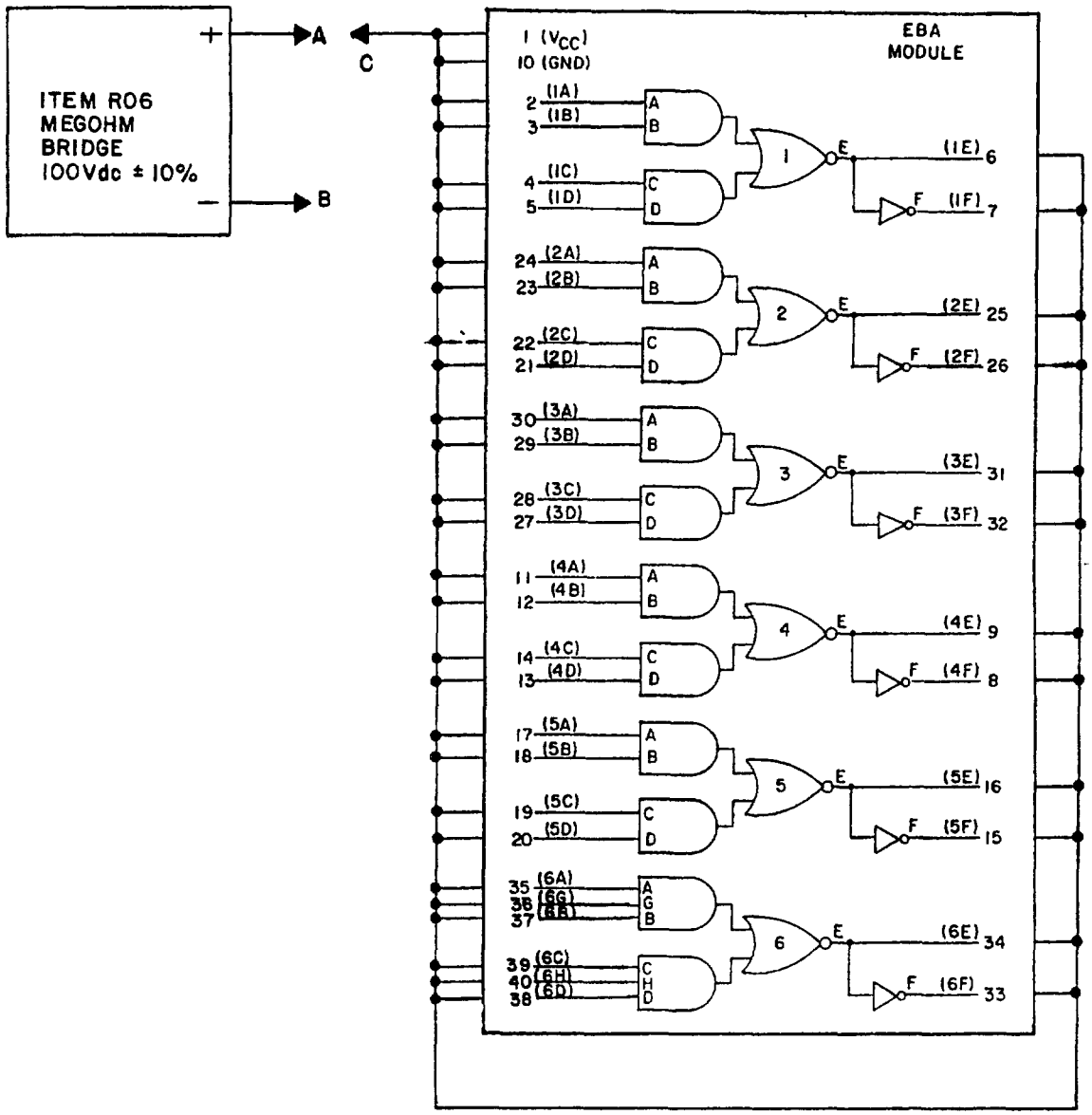
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation, frame to module pins	C	Frame	10	∞	MΩ	7

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WARNING: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Output voltage (low-level) test.

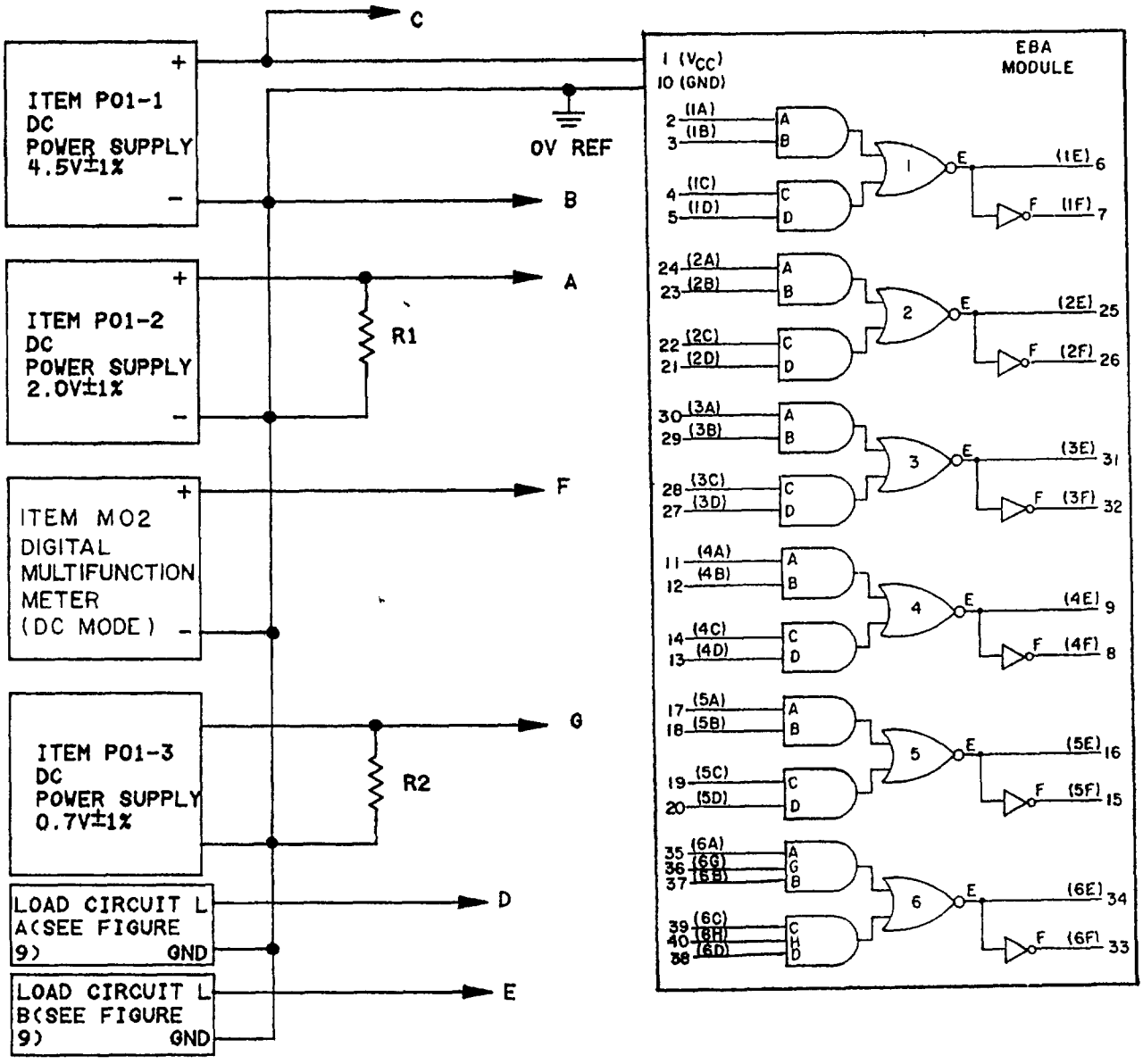
Test condition	Parameter	Connect							Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
D1	1E Output voltage, low level	2,3	4,5	Open	7	6	6	Open	0.0	0.3	V	8
D2	1E	4,5	2,3		7	6	6					
D3	2E	24,23	22,21		26	25	25					
D4	2E	22,21	24,23		26	25	25					
D5	3E	30,29	28,27		32	31	31					
D6	3E	28,27	30,29		32	31	31					
D7	4E	12,11	14,13		8	9	9					
D8	4E	14,13	12,11		8	9	9					
D9	5E	17,18	20,19		15	16	16					
D10	5E	20,19	17,18		15	16	16					
D11	6E	35, 36,37	38, 39,40		33	34	34					
D12	6E	38, 39,40	35, 36,37		33	34	34					
D13	1F	Open	Open	3,5	7	6	7	2,4				
D14	2F			21,23	26	25	26	22,24				
D15	3F			27,29	32	31	32	28,30				
D16	4F			11,13	8	9	8	12,14				
D17	5F			18,19	15	16	15	17,20				
D18	6F			36,37, 38,39	33	34	33	35,40				

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TABLE VII. Output voltage (high-level) test.

Test condition	Parameter	Connect							Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
E1	1E Output voltage, high level	Open	Open	3,5	7	6	6	2,4	2.4	4.5	V	8
E2	1E			2,4	7	6	6	3,5				
E3	2E			21,23	26	25	25	22,24				
E4	2E			22,24	26	25	25	21,23				
E5	3E			27,29	32	31	31	28,30				
E6	3E			28,30	32	31	31	27,29				
E7	4E			11,13	8	9	9	12,14				
E8	4E			12,14	8	9	9	11,13				
E9	5E			18,19	15	16	16	17,20				
E10	5E			17,20	15	16	16	18,19				
E11	6E			36,37, 39,40	33	34	34	35,38				
E12	6E			35,36, 38,40	33	34	34	37,39				
E13	1F	2,3	4,5	Open	7	6	7	Open				
E14	2F	23,24	21,22		26	25	26					
E15	3F	29,30	27,28		32	31	32					
E16	4F	11,12	13,14		8	9	8					
E17	5F	17,18	19,20		15	16	15					
E18	6F	35, 36,37	38, 39,40		33	34	33					

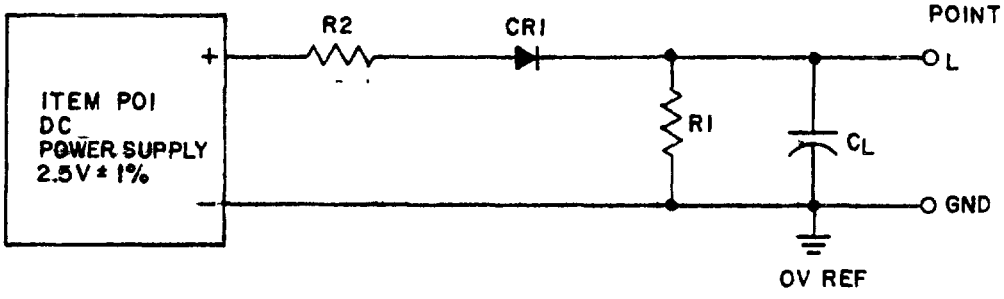
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NOTES.
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Both outputs of the circuit under test shall be loaded simultaneously.
3. Since the items P01-2 and P01-3 are not used simultaneously, only two item P01 supplies are required.

FIGURE 8 Output voltage (low-level and high-level) test circuit for tables VI and VII.

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NOTES. LOAD CIRCUIT "A" (for F outputs):

1. CR1 = 1N916 or equivalent.
2. R1 = 23.7 kΩ ±1%, 1/8W with a current flow of 100 µA when point L is held at 2.4 V.
3. R2 = Select value for a current flow of 2.0 mA ±5% out of point L when point L is held at 0.3 V.
4. C_L = 40 ±5pF (including probe capacitance) for test conditions K1-K12, L1-L12, M1-M12 and N1-N12.
   = 140 ±5pF (including probe capacitance) for test conditions K13-K24, L13-L24, M13-M24 and N13-N24
   = 50 ±5pF (including probe capacitance) for test conditions P1-P12 and R1-R12.
5. C_L is not required for DC electrical tests.
6. The maximum lead length from point "L" to the appropriate module pin shall be less than 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.
8. See figure 1 for definition of module E and F outputs

LOAD CIRCUIT "B" (for E outputs):

1. Same as notes 1, 4, 5, 6, 7, 8 for load circuit A.
2. R1 = 26.7 kΩ ±1%, 1/8W with a current flow of 100 µA when point L is held at 2.4 V.
3. R2 = Select value for a current flow of 1.8 mA ±5% out of point L when point L is held at 0.3 V.

FIGURE 9. EBA logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1A Input current, low level	Open	2	Open	3	50	200	μA	10
F2	1B		3		2				
F3	1C		4		5				
F4	1D		5		4				
F5	2A		24		23				
F6	2B		23		24				
F7	2C		22		21				
F8	2D		21		22				
F9	3A		30		29				
F10	3B		29		30				
F11	3C		28		27				
F12	3D		27		28				
F13	4B		12		11				
F14	4A		11		12				
F15	4C		14		13				
F16	4D		13		14				
F17	5A		17		18				
F18	5B		18		17				
F19	5C		19		20				
F20	5D		20		19				
F21	6A		35		36,37				
F22	6G		36		37,35				
F23	6B		37		35,36				
F24	6D		38		39,40				
F25	6C		39		38,40				
F26	6H		40		38,39				

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TABLE IX. Input current (high-level) test.

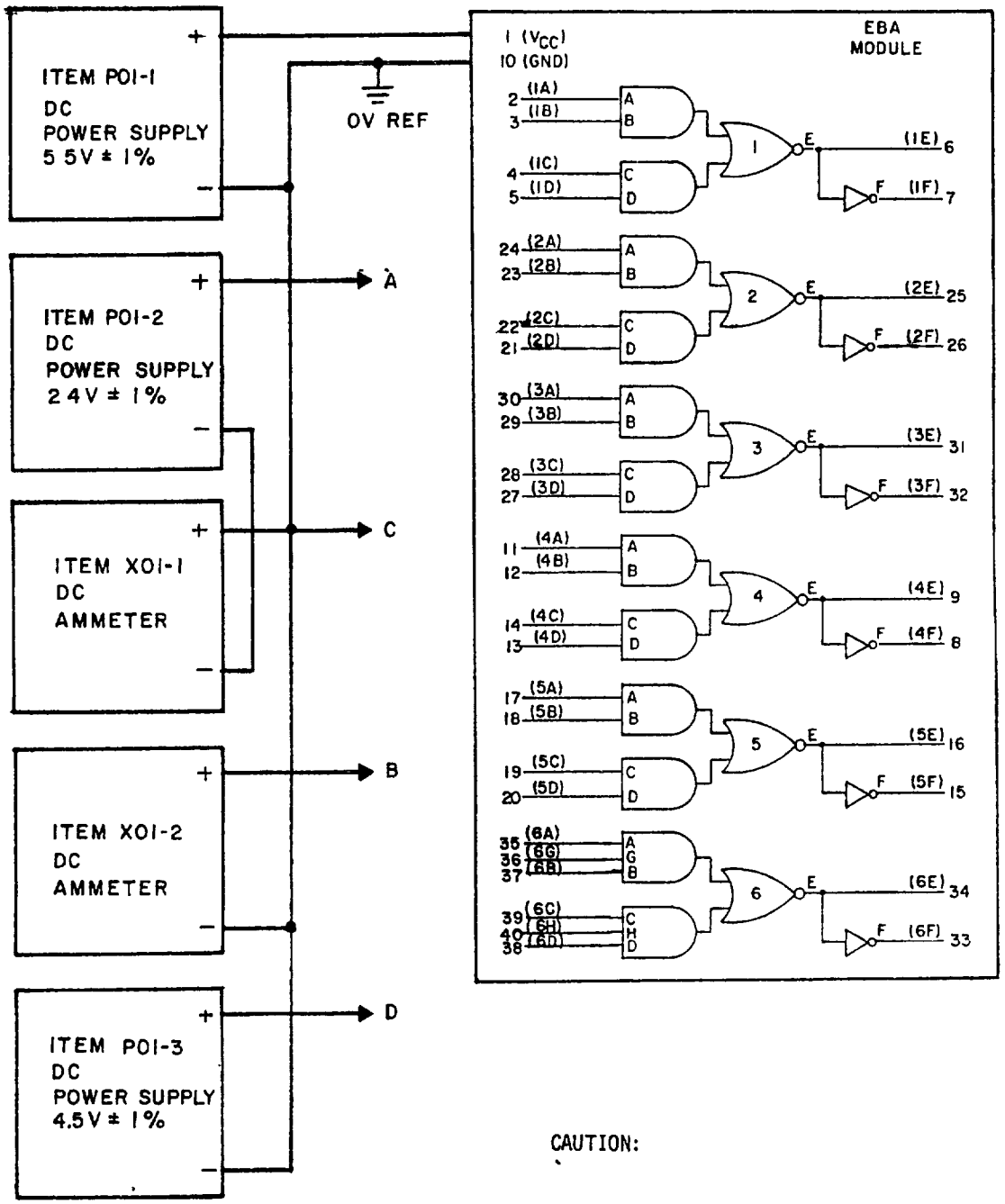
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1A Input current, high level	2	Open	3	Open	0	10	µA	10
G2	1B	3		2					
G3	1C	4		5					
G4	1D	5		4					
G5	2A	24		23					
G6	2B	23		24					
G7	2C	22		21					
G8	2D	21		22					
G9	3A	30		29					
G10	3B	29		30					
G11	3C	28		27					
G12	3D	27		28					
G13	4B	12		11					
G14	4A	11		12					
G15	4C	14		13					
G16	4D	13		14					
G17	5A	17		18					
G18	5B	18		17					
G19	5C	19		20					
G20	5D	20		19					
G21	6A	35		36,37					
G22	6G	36		37,35					
G23	6B	37		35,36					
G24	6D	38		39,40					
G25	6C	39		38,40					
G26	6H	40		38,39					

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1E Output short circuit current	Open	6	2,3,4,5	Open	3	15	mA	10
H2	2E		25	21,22, 23,24					
H3	3E		31	27,28, 29,30					
H4	4E		9	11,12, 13,14					
H5	5E		16	17,18, 19,20					
H6	6E		34	35,36,37, 38,39,40					
H7	1F		7	Open	2,3				
H8	2F		26		23,24				
H9	3F		32		29,30				
H10	4F		8		11,12				
H11	5F		15		17,18				
H12	6F		33		35,36,37				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. Since items P01-2 and P01-3 are not used simultaneously, only two item P01 supplies are required.
3. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.
4. When performing output short circuit test, only one output shall be shorted at a time and the measurement taken within 5 seconds.

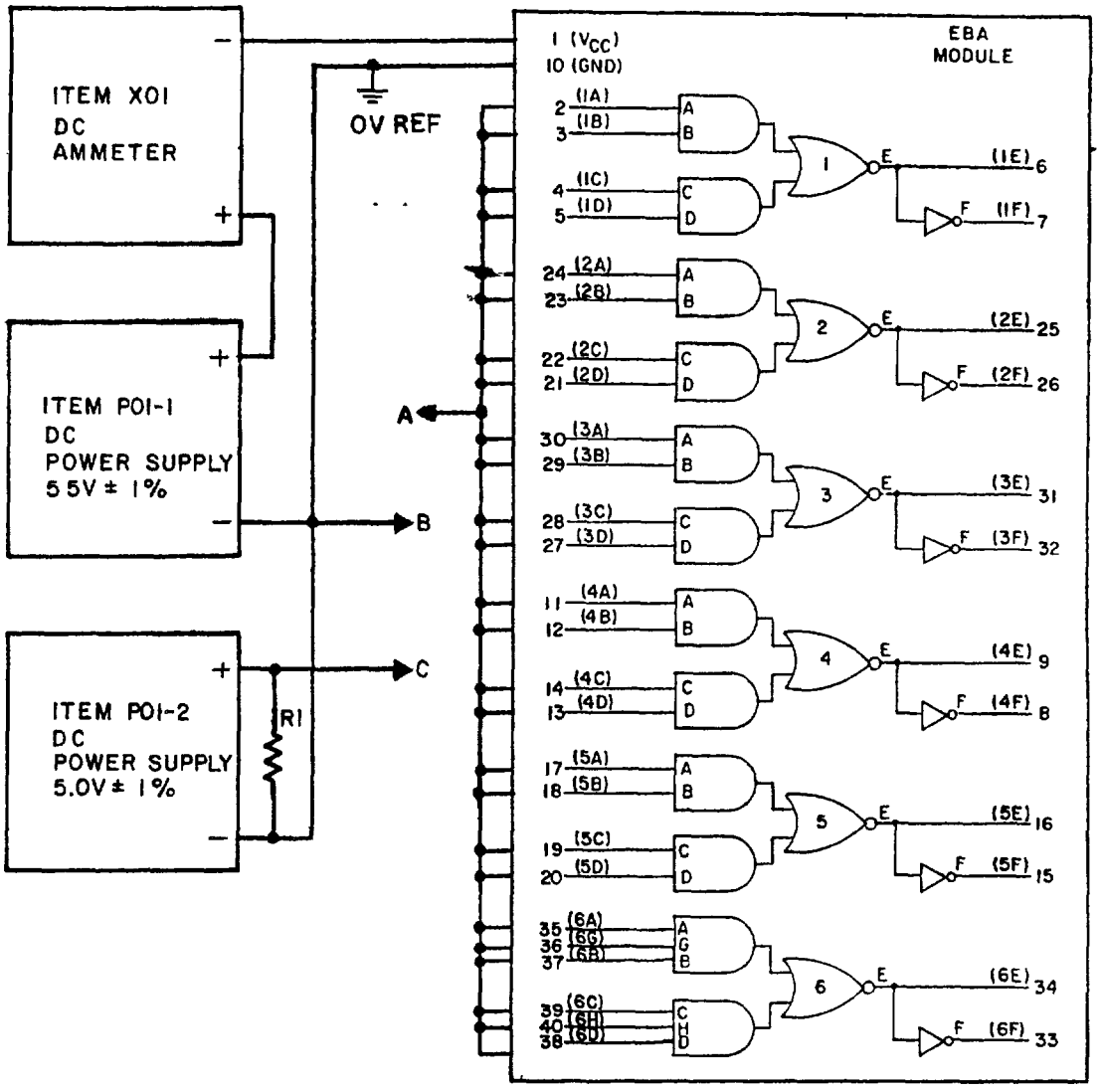
FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Power supply current, low level	C	Open	A	0.5	5.1	mA	11
J2	Power supply current, high level	B	A	Open	0.2	5.5	mA	11

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: R1 = 51 Ω ±5%, 1.W.

FIGURE 11. Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
K1	1E Transition time (t_{TLH}), C_L = 40 pF	2	3	4	7	6	6	5	60	ns	12, 13
K2	2E	24	23	22	26	25	25				
K3	3E	30	29	28	32	31	31				
K4	4E	12	11	14	8	9	9				
K5	5E	17	18	19	15	16	16				
K6	6E	35	37,36	38	33	34	34				
K7	1F	2	3	4	7	6	7				
K8	2F	24	23	22	26	25	26				
K9	3F	30	29	28	32	31	32				
K10	4F	12	11	14	8	9	8				
K11	5F	17	18	19	15	16	15				
K12	6F	35	36,37	38	33	34	33				
K13	1E Transition time (t_{TLH}), C_L = 140 pF	2	3	4	7	6	6	10	100		
K14	2E	24	23	22	26	25	25				
K15	3E	30	29	28	32	31	31				
K16	4E	12	11	14	8	9	9				
K17	5E	17	18	19	15	16	16				
K18	6E	35	37,36	38	33	34	34				
K19	1F	2	3	4	7	6	7				
K20	2F	24	23	22	26	25	26				
K21	3F	30	29	28	32	31	32				
K22	4F	12	11	14	8	9	8				
K23	5F	17	18	19	15	16	15				
K24	6F	35	36,37	38	33	34	33				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
L1	1E Transition time (t_{THL}), C_L = 40 pF	2	3	4	7	6	6	2	30	ns	12, 13
L2	2E	24	23	22	26	25	25				
L3	3E	30	29	28	32	31	31				
L4	4E	12	11	14	8	9	9				
L5	5E	17	18	19	15	16	16				
L6	6E	35	37,36	38	33	34	34				
L7	1F	2	3	4	7	6	7				
L8	2F	24	23	22	26	25	26				
L9	3F	30	29	28	32	31	32				
L10	4F	12	11	14	8	9	8				
L11	5F	17	18	19	15	16	15				
L12	6F	35	36,37	38	33	34	33				
L13	1E Transition time (t_{THL}), C_L = 140 pF	2	3	4	7	6	6	5	60		
L14	2E	24	23	22	26	25	25				
L15	3E	30	29	28	32	31	31				
L16	4E	12	11	14	8	9	9				
L17	5E	17	18	19	15	16	16				
L18	6E	35	37,36	38	33	34	34				
L19	1F	2	3	4	7	6	7				
L20	2F	24	23	22	26	25	26				
L21	3F	30	29	28	32	31	32				
L22	4F	12	11	14	8	9	8				
L23	5F	17	18	19	15	16	15				
L24	6F	35	36,37	38	33	34	33				

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TABLE XIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M1	1E Propagation delay time t_{PLH}, C_L = 40 pF	2	3	4,5	7	6	6	10	90	ns	12, 13
M2	1E	4	5	2,3	7	6	6				
M3	2E	24	23	21,22	26	25	25				
M4	2E	22	21	23,24	26	25	25				
M5	3E	30	29	27,28	32	31	31				
M6	3E	28	27	29,30	32	31	31				
M7	4E	12	11	13,14	8	9	9				
M8	4E	14	13	11,12	8	9	9				
M9	5E	17	18	19,20	15	16	16				
M10	5E	20	19	17,18	15	16	16				
M11	6E	35	36,37	38, 39,40	33	34	34				
M12	6E	38	40,39	35, 36,37	33	34	34				
M13	1E Propagation delay time, t_{PLH}, C_L = 140 pF	2	3	4,5	7	6	6		120		
M14	1E	4	5	2,3	7	6	6				
M15	2E	24	23	21,22	26	25	25				
M16	2E	22	21	23,24	26	25	25				
M17	3E	30	29	27,28	32	31	31				
M18	3E	28	27	29,30	32	31	31				
M19	4E	12	11	13,14	8	9	9				
M20	4E	14	13	11,12	8	9	9				
M21	5E	17	18	19,20	15	16	16				
M22	5E	20	19	17,18	15	16	16				
M23	6E	35	36,37	38, 39,40	33	34	34				
M24	6E	38	40,39	35, 36,37	33	34	34				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
N1	1F Propagation delay time, t_{PLH}, C_L = 40 pF	2	3	4,5	7	6	7	10	150	ns	12, 13
N2	1F	4	5	2,3	7	6	7				
N3	2F	24	23	21,22	26	25	26				
N4	2F	22	21	23,24	26	25	26				
N5	3F	30	29	27,28	32	31	32				
N6	3F	28	27	29,30	32	31	32				
N7	4F	12	11	13,14	8	9	8				
N8	4F	14	13	11,12	8	9	8				
N9	5F	17	18	19,20	15	16	15				
N10	5F	20	19	17,18	15	16	15				
N11	6F	35	36,37	38, 39,40	33	34	33				
N12	6F	38	40,39	35, 36,37	33	34	33				
N13	1F Propagation delay time, t_{PLH}, C_L = 140 pF	2	3	4,5	7	6	7		180		
N14	1F	4	5	2,3	7	6	7				
N15	2F	24	23	21,22	26	25	26				
N16	2F	22	21	23,24	26	25	26				
N17	3F	30	29	27,28	32	31	32				
N18	3F	28	27	29,30	32	31	32				
N19	4F	12	11	13,14	8	9	8				
N20	4F	14	13	11,12	8	9	8				
N21	5F	17	18	19,20	15	16	15				
N22	5F	20	19	17,18	15	16	15				
N23	6F	35	36,37	38, 39,40	33	34	33				
N24	6F	38	40,39	35, 36,37	33	34	33				

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TABLE XV. Propagation delay time (t_{PHL}) test.

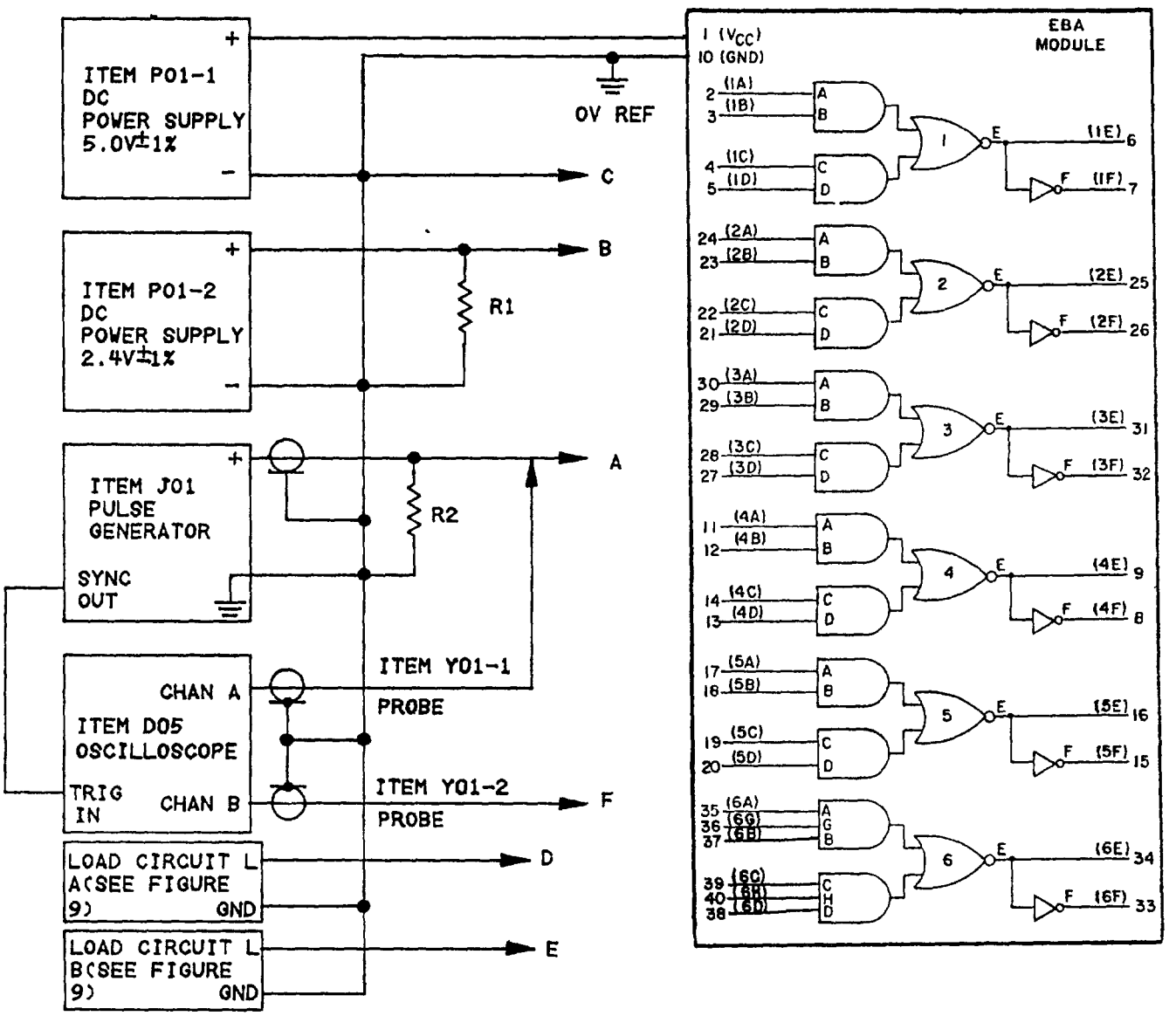
Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
P1	1E Propagation delay time, t_{PHL}, C_L = 40 pF	2	3	4,5	7	6	6	10	60	ns	12, 13
P2	1E	4	5	2,3	7	6	6				
P3	2E	24	23	21,22	26	25	25				
P4	2E	22	21	23,24	26	25	25				
P5	3E	30	29	27,28	32	31	31				
P6	3E	28	27	29,30	32	31	31				
P7	4E	12	11	13,14	8	9	9				
P8	4E	14	13	11,12	8	9	9				
P9	5E	17	18	19,20	15	16	16				
P10	5E	20	19	17,18	15	16	16				
P11	6E	35	36,37	38, 39,40	33	34	34				
P12	6E	38	40,39	35, 36,37	33	34	34				
P13	1E Propagation delay time, t_{PHL}, C_L = 140 pF	2	3	4,5	7	6	6		90		
P14	1E	4	5	2,3	7	6	6				
P15	2E	24	23	21,22	26	25	25				
P16	2E	22	21	23,24	26	25	25				
P17	3E	30	29	27,28	32	31	31				
P18	3E	28	27	29,30	32	31	31				
P19	4E	12	11	13,14	8	9	9				
P20	4E	14	13	11,12	8	9	9				
P21	5E	17	18	19,20	15	16	16				
P22	5E	20	19	17,18	15	16	16				
P23	6E	35	36,37	38, 39,40	33	34	34				
P24	6E	38	40,39	35, 36,37	33	34	34				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R1	1F Propagation delay time, t_{PHL}, C_L = 40 pF	2	3	4,5	7	6	7	10	120	ns	12, 13
R2	1F	4	5	2,3	7	6	7				
R3	2F	24	23	21,22	26	25	26				
R4	2F	22	21	23,24	26	25	26				
R5	3F	30	29	27,28	32	31	32				
R6	3F	28	27	29,30	32	31	32				
R7	4F	12	11	13,14	8	9	8				
R8	4F	14	13	11,12	8	9	8				
R9	5F	17	18	19,20	15	16	15				
R10	5F	20	19	17,18	15	16	15				
R11	6F	35	36,37	38, 39,40	33	34	33				
R12	6F	38	40,39	35, 36,37	33	34	33				
R13	1F Propagation delay time, t_{PHL}, C_L = 140 pF	2	3	4,5	7	6	7		150		
R14	1F	4	5	2,3	7	6	7				
R15	2F	24	23	21,22	26	25	26				
R16	2F	22	21	23,24	26	25	26				
R17	3F	30	29	27,28	32	31	32				
R18	3F	28	27	29,30	32	31	32				
R19	4F	12	11	13,14	8	9	8				
R20	4F	14	13	11,12	8	9	8				
R21	5F	17	18	19,20	15	16	15				
R22	5F	20	19	17,18	15	16	15				
R23	6F	35	36,37	38, 39,40	33	34	33				
R24	6F	38	40,39	35, 36,37	33	34	33				

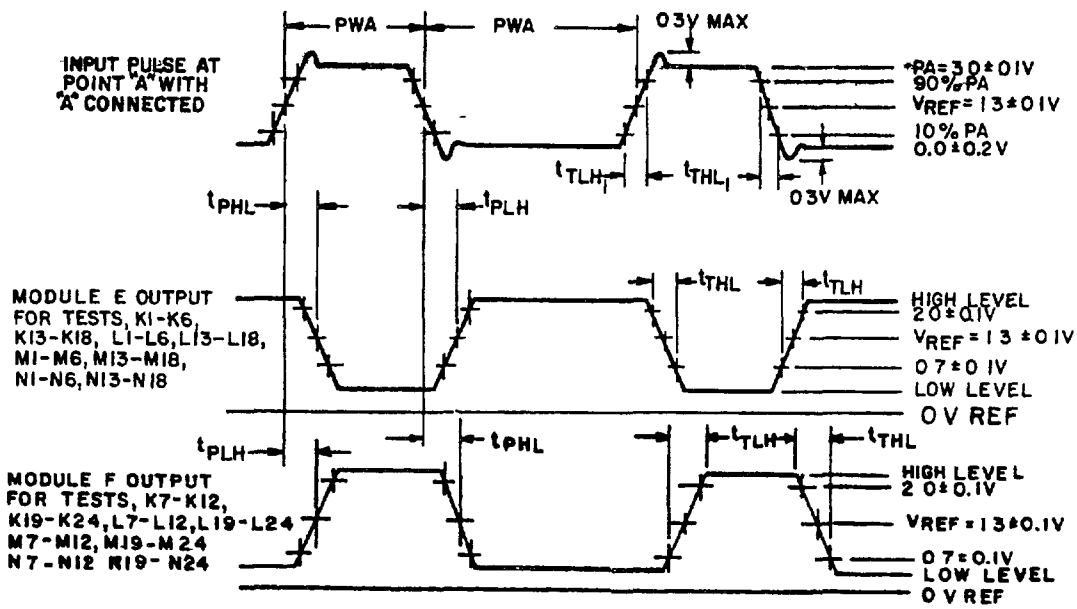
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Both outputs of the circuit under test shall be loaded simultaneously

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII, XIV, and XV.

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NOTES:

1. Input pulse parameters:

   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns

   PWA = 500 ±50 ns

2. Before making the measurements verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.
3. All voltages are referenced to ground.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII, XIV and XV.

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TABLE XVI Output crosstalk (low-level) test.

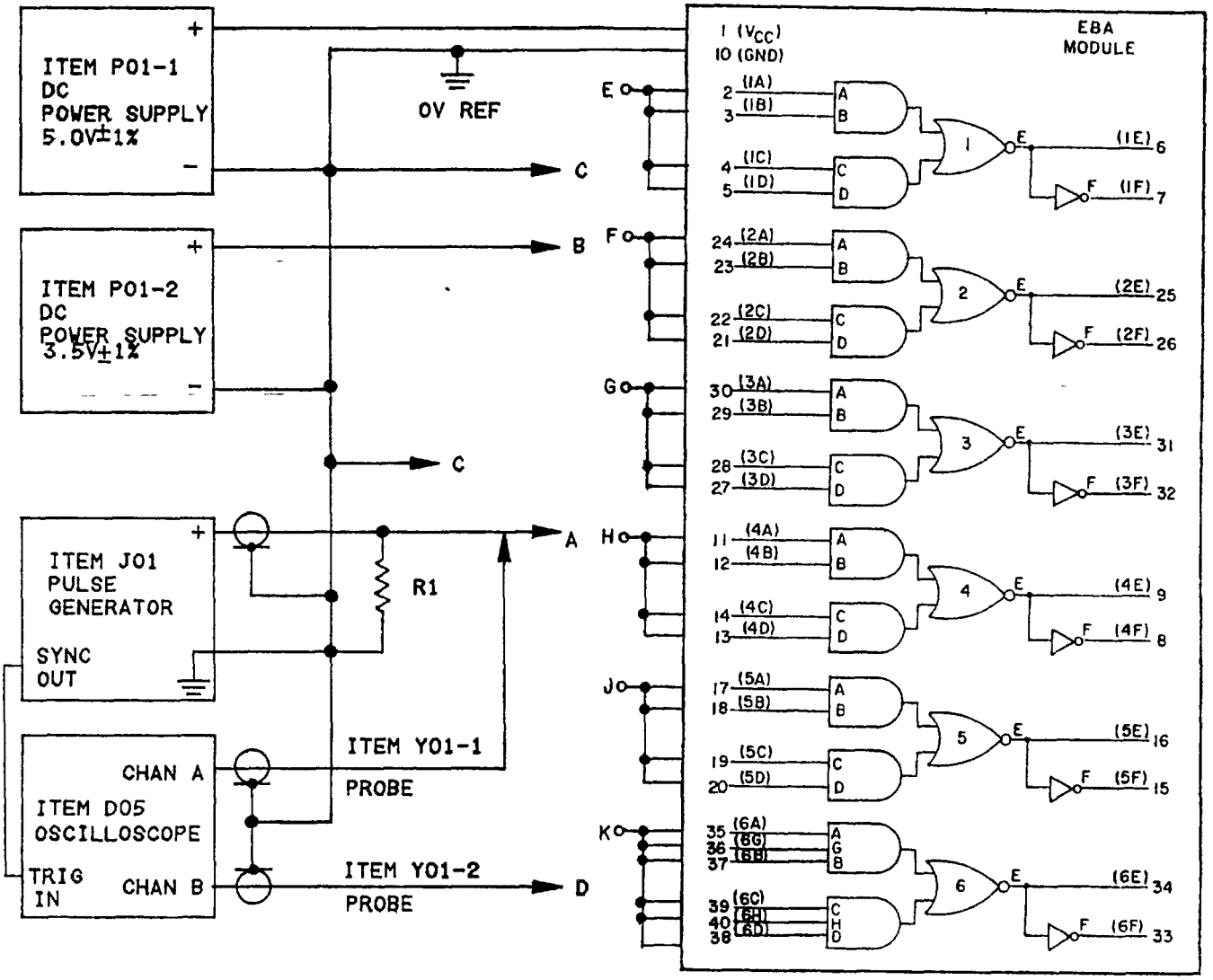
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1E Output crosstalk, low level	F,G,H,J,K	E	Open	6	0	400	mV	14, 15
S2	1F	F,G,H,J,K	Open	E	7				
S3	2E	E,G,H,J,K	F	Open	25				
S4	2F	E,G,H,J,K	Open	F	26				
S5	3E	E,F,H,J,K	G	Open	31				
S6	3F	E,F,H,J,K	Open	G	32				
S7	4E	E,F,G,J,K	H	Open	9				
S8	4F	E,F,G,J,K	Open	H	8				
S9	5E	E,F,G,H,K	J	Open	16				
S10	5F	E,F,G,H,K	Open	J	15				
S11	6E	E,F,G,H,J	K	Open	34				
S12	6F	E,F,G,H,J	Open	K	33				

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1E Output crosstalk, high level	F,G,H,J,K	Open	E	6	0	400	mV	14, 15
T2	1F	F,G,H,J,K	E	Open	7				
T3	2E	E,G,H,J,K	Open	F	25				
T4	2F	E,G,H,J,K	F	Open	26				
T5	3E	E,F,H,J,K	Open	G	31				
T6	3F	E,F,H,J,K	G	Open	32				
T7	4E	E,F,G,J,K	Open	H	9				
T8	4F	E,F,G,J,K	H	Open	8				
T9	5E	E,F,G,H,K	Open	J	16				
T10	5F	E,F,G,H,K	J	Open	15				
T11	6E	E,F,G,H,J	Open	K	34				
T12	6F	E,F,G,H,J	K	Open	33				

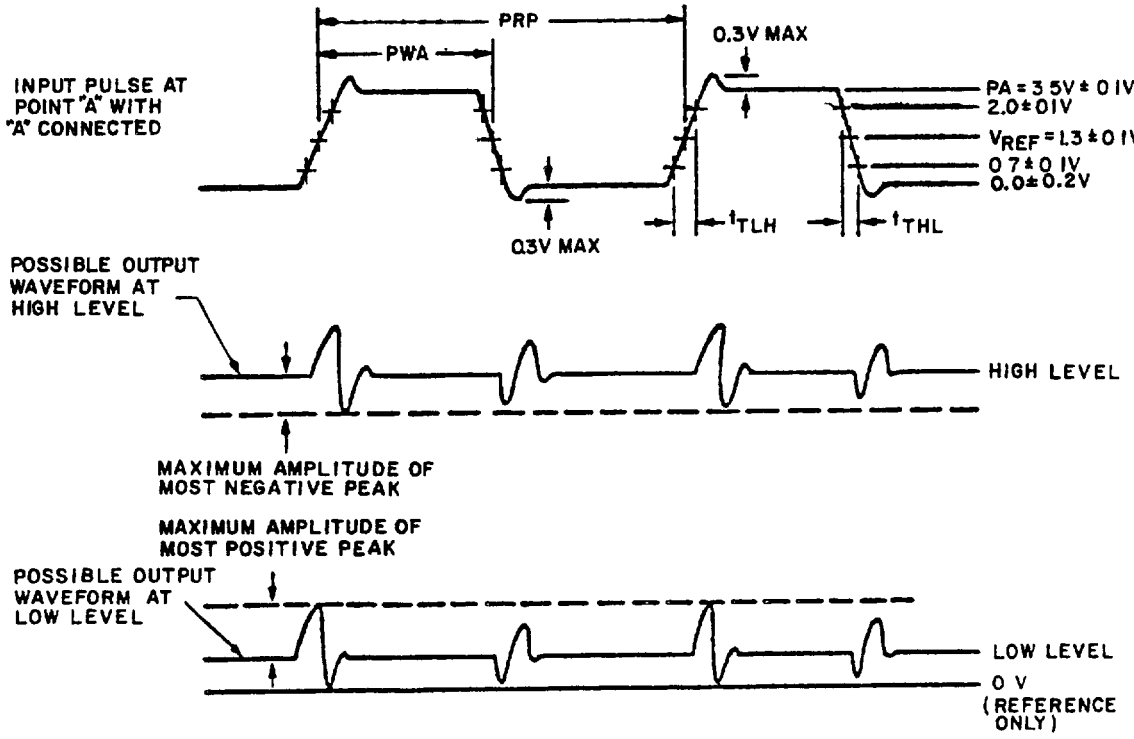
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 9.

FIGURE 14 Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all inputs)
   PWA = 200 ±20 ns
   PRP = 1,000 ±100 ns
2. All voltages are referenced to ground.

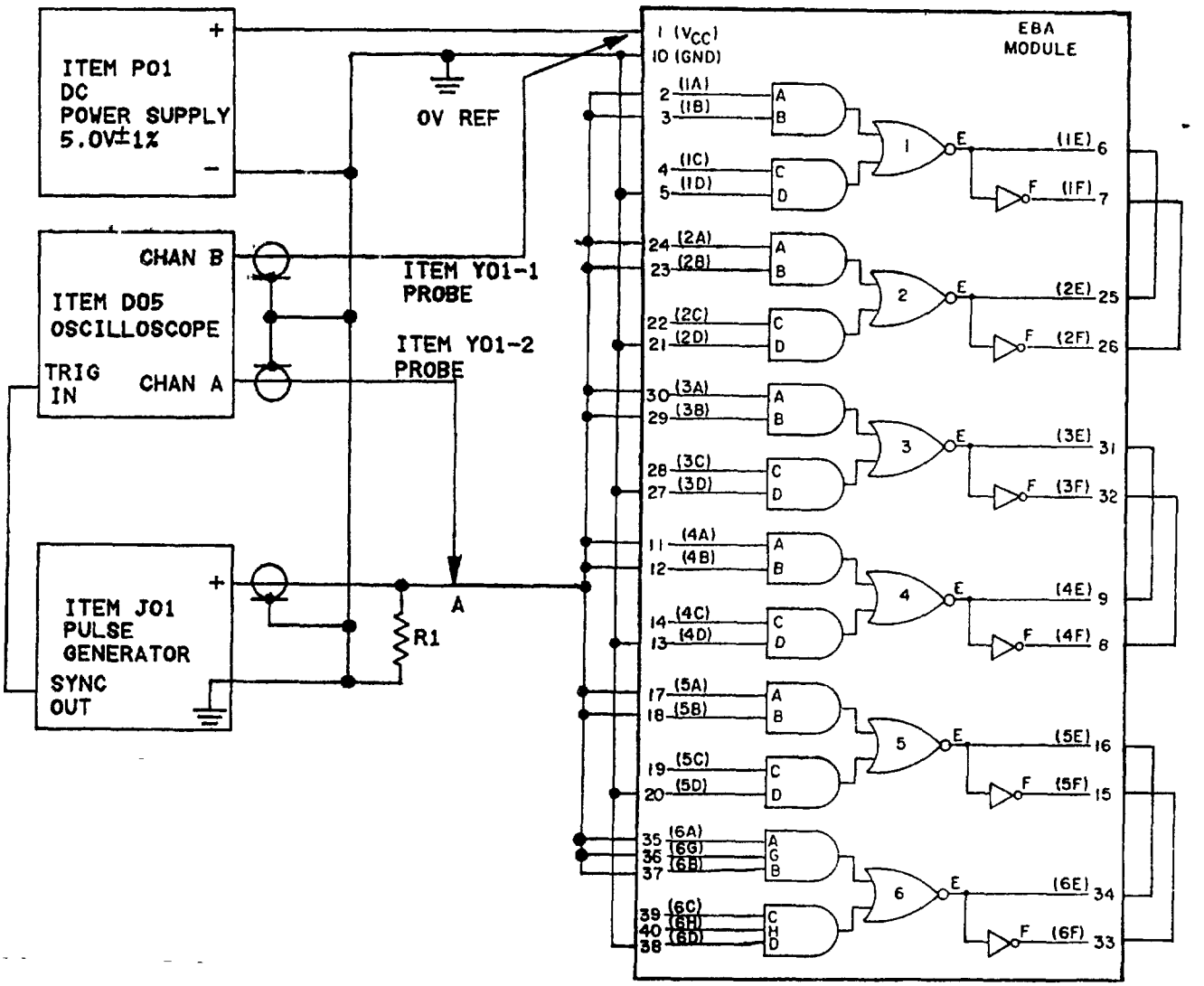
FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	300	mV	16, 17

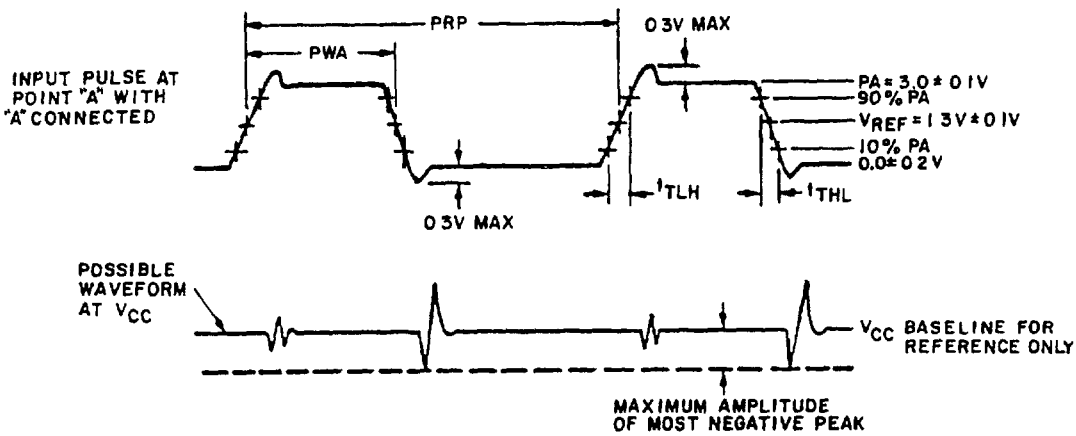
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 16. Power supply transient amplitude test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 20 \pm 2$ ns
   PWA = 200 ±20 ns
   PRP = 1,000 ±100 ns
2. All voltages are referenced to ground.

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX Power supply filter capacitance test.

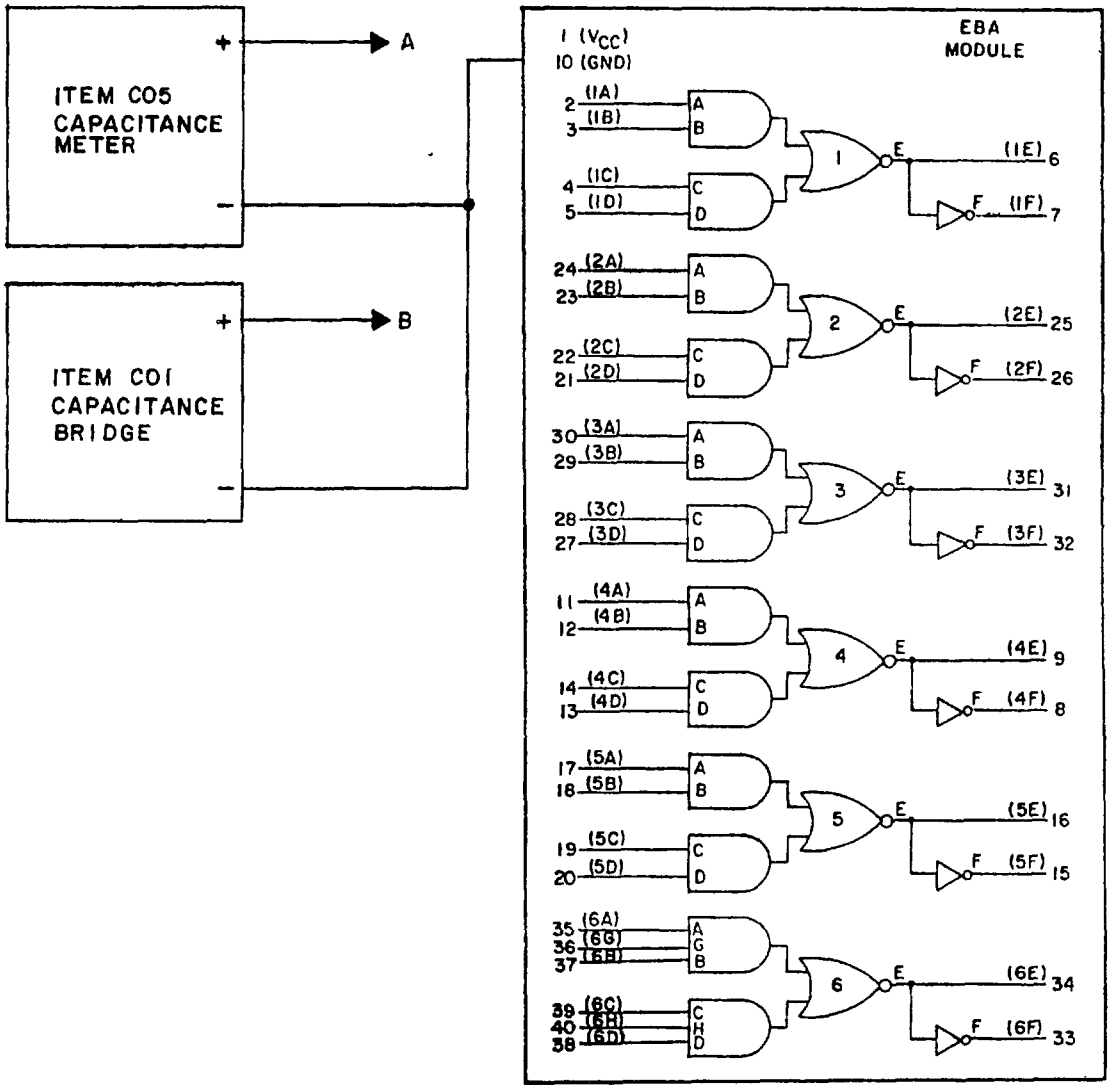
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1 65	2.75	µF	18

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
W1	1A Input capacitance	2	2	15	pF	18
W2	1B	3				
W3	1C	4				
W4	1D	5				
W5	2A	24				
W6	2B	23				
W7	2C	22				
W8	2D	21				
W9	3A	30				
W10	3B	29				
W11	3C	28				
W12	3D	27				
W13	4B	12				
W14	4A	11				
W15	4C	14				
W16	4D	13				
W17	5A	17				
W18	5B	18				
W19	5D	20				
W20	5C	19				
W21	6A	35				
W22	6B	37				
W23	6D	38				
W24	6C	39				
W25	6G	36				
W26	6H	40				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V max.

FIGURE 18. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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